Technical Report – Point D

Demonstration of Detection and Avoidance Architectures

for Safe Integration of Unmanned Aircraft System (UAS) in

Non-Segregated Airspace

SIMUA Project – *Safe Integration of Manned and Unmanned Aircrafts*

Brazil-Sweden Strategic Partnership | Air Domain Study (ADS)

Instituto Tecnológico de Aeronáutica (ITA)
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March 2026

Abstract

This report documents the technical results of **Point D** of the Work Plan for the SIMUA project (*Safe Integration of Manned and Unmanned Aircrafts*), developed within the framework of the strategic partnership between Instituto Tecnológico de Aeronáutica (ITA) and Instituto de Controle do Espaço Aéreo (ICEA) with the Swedish *Air Domain Study* (ADS) program. The document presents the architecture selection methodology for Detect and Avoid (DAA) systems in non-segregated airspaces, structured around the international Remain Well Clear (RWC)/Collision Avoidance (CA) layered framework (RTCA DO-365B, ASTM F3442). The work covers: (i) the trade-off study of candidate sensors (LiDAR, mmWave Radar, Computer Vision, and Acoustics) with justification of Size, Weight and Power (SWaP) constraints; (ii) the definition of detection envelopes and Operational Services and Environment Definition (OSED) mapped to Brazilian Airspace Control System (SISCEAB) airspace classes (G, E, D, C); (iii) integration with the DAIDALUS algorithm as the RWC implementation, including Sensor Uncertainty Mitigation (SUM)/Virtual Intruder State Aggregation (VISA) logic for non-cooperative scenarios and Human-in-the-Loop (HITL) alerting design; and (iv) area saturation

analysis validating operational capacity (> 7 UAS/km^2) in dense urban environments. The experimental campaign is organized into two complementary scales: Macro-Scale Strategic Layering Analysis (MSLA) for strategic UTM/DAA layering validation via encounter models, and High-Fidelity Tactical Logic Validation (HTLV) for high-fidelity tactical logic validation at 50 Hz kinematic updates. The results demonstrate that the hybrid Airborne Detect and Avoid (ABDAA) architecture, combining mmWave Radar and Computer Vision with DAIDALUS SUM logic, is the most effective solution for ensuring safety and scalability of drone traffic in Brazilian airspace.

Keywords: Detect and Avoid; Remain Well Clear; Collision Avoidance; UAS; DAIDALUS; mmWave Radar; Computer Vision; UTM; OSED; Encounter Models; Risk Ratio; SIMUA; SWaP; DO-365B; ASTM F3442.

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1 Technical Foundation and Requirements

This section establishes the technical and strategic basis for the SIMUA project's Point D milestone. It bridges the international cooperation goals with the physical engineering requirements for safe UAS integration in non-segregated airspace, covering the sensor architecture trade-offs and the operational detection envelopes.

1.1 Strategic Alignment and Partnership

1.1.1 Context and Motivation

The integration of remotely piloted aircraft (UAS) into shared airspaces with manned aviation is one of the fundamental milestones for the feasibility of Advanced Air Mobility (Advanced Air Mobility (AAM)) and Unmanned Aircraft Systems Traffic Management (Unmanned Traffic Management (UTM)). Within the scope of the strategic partnership between ITA and ICEA, the SIMUA project establishes a rigorous scientific framework for the transition from simple flight tests to a robust methodology for technological architecture selection.

The complexity of this initiative is reflected in the Brazil-Sweden international cooperation, specifically in the Air Domain Study (ADS) program, where the increased scope to include analyses in controlled airspace and AAM flows requires a DAA architecture that is simultaneously resource-efficient and highly reliable. The transition from cooperative, telemetry-based systems to hybrid systems capable of detecting non-cooperative aircraft represents the core of the technological development detailed here.

1.1.2 DAA Layered Architecture: RWC and CA

In alignment with international DAA standards [1, 2], this report adopts a two-layer separation model:

- **RWC**: The tactical self-separation function that maintains the Well Clear volume between aircraft. In this report, the DAIDALUS algorithm is the primary implementation of the RWC layer, providing multi-level alerting (Preventive, Corrective, Warning) and maneuver guidance bands to the Remote Pilot in Command (RPIC).
- **CA**: The emergency last-resort maneuver layer activated only when the RWC function has failed to prevent a loss of Well Clear. CA logic triggers aggressive escape maneuvers to prevent a Near Mid-Air Collision (NMAC).

This distinction ensures that the report aligns with the RWC/CA framework defined in RTCA DO-365B and ASTM F3442/F3442M.

1.1.3 Operational Services and Environment Definition (OSED)

The OSED for this study maps simulations to specific airspace classes as defined in the SISCEAB framework:

- **Class G** (Uncontrolled): Primary environment for initial Beyond Visual Line of Sight (BVLOS) UAS operations; no ATC separation service is provided. DAA self-separation is the sole barrier against collision.
- **Class E** (Controlled, IFR only): Transition environment where UAS must maintain self-separation from VFR traffic while complying with IFR clearances.
- **Classes C and D** (Controlled, IFR + VFR): High-complexity environments requiring full DAA capability with Automatic Dependent Surveillance-Broadcast (ADS-B) cooperative surveillance, supplemented by onboard sensors for non-cooperative intruders.

The simulation campaigns (MSLA and HTLV) are designed to validate DAA performance across these operational envelopes, with increasing complexity from Class G through Class C.

1.1.4 Milestone Identification – Point D

Point D of the ITA/ICEA Work Plan corresponds to the **Architecture Demonstration** stage, comprising:

1. The validation of drone integration in non-segregated airspace through DAA technologies, with explicit RWC and CA layers as defined in DO-365B [1];
2. The demonstration of the effectiveness of the DAIDALUS algorithm as an implementation of the RWC standard in cooperative scenarios (via ADS-B/telemetry) and non-cooperative scenarios (via onboard sensors);
3. The proof of concept of the hybrid ABDAA architecture as a scalable solution for high-density UTM environments, benchmarked against ASTM F3442 performance requirements for Small Unmanned Aircraft System (sUAS) [2].

1.1.5 Partners and Responsibilities

The following institutional actors participate in the execution of Point D activities:

- **ITA**: Technical leadership in the development of simulations, sensor modeling, and integration with DAIDALUS.

- **ICEA**: Regulatory expertise (SISCEAB/Departamento de Controle do Espaço Aéreo (DECEA)) and provision of the national operational scenario.
- **Saab (Sweden)**: Technological support in surveillance systems and sensor fusion algorithms.
- **Luftfartsverket (Swedish Civil Aviation Authority) (LFV)**: European normative reference for controlled airspace and UTM.

1.2 Architecture Definition: Sensors and Envelopes

1.2.1 Trade-off Study: Motivation and Criteria

The selection of sensors for ABDAA systems is governed by SWaP constraints and the performance-based framework defined in the SIMUA Concept of Operations (CONOPS). These requirements are particularly critical for small drones – such as the DJI Mini line or custom platforms for urban delivery – which represent the majority of operations in the high-density urban corridors envisioned for the project. The engineering challenge lies in balancing the need for high spatial resolution with the limited payload capacity and power budget of the aircraft (*ownship*).

The criteria adopted for the *trade-off* include:

- Power consumption (Watts);
- Module weight (grams);
- Computational processing cost;
- Detection latency (ms);
- Resilience to adverse weather conditions;
- Typical detection range for UAS.

1.2.2 LiDAR: Evaluation and Infeasibility on Micro-UAS

While LiDAR (*Light Detection and Ranging*) is widely recognized for its centimeter precision and ability to generate detailed 3D maps, its application as a primary DAA sensor on small drones faces significant physical barriers. High-performance units, such as the Ouster OS0, offer a 90° vertical field of view and up to 128 resolution channels, but operate with power consumption between 14 and 20 W, reaching peaks of 28 W during startup or extreme temperature conditions [3].

For a platform whose flight autonomy is dictated by grams of battery, the weight of approximately 430 g of the OS0 sensor, combined with the need for computational

processing to handle millions of points per second, makes the system infeasible for micro-drones [4]. Furthermore, LiDAR exhibits severe degradation in adverse atmospheric conditions, such as dense fog, smoke, or heavy rain, where laser pulses suffer scattering, drastically reducing the probability of detecting intruder aircraft at long distances [5].

1.2.3 Millimeter Wave Radar (mmWave)

mmWave Radar, exemplified by the Texas Instruments IWR6843 line, operates in the 60 GHz to 64 GHz band, offering an extremely compact and low-power solution (< 2–5 W) [6]. Frequency Modulated Continuous Wave (FMCW) technology allows the radar to extract not only distance but also relative velocity and angle of arrival (AoA) of multiple targets simultaneously, with high immunity to weather and lighting conditions [7].

1.2.4 Computer Vision with Edge AI

Computer Vision leverages the ubiquity of lightweight cameras and the advancement of hardware accelerators such as the NVIDIA Jetson Orin Nano and Google Coral Edge TPU [8]. Neural networks like YOLOv8 and YOLOv10 allow real-time detection, although they require high processing costs (measured in TOPS or MFLOPS) [9, 10].

1.2.5 Passive Acoustic Array

Passive acoustic systems, based on microphone arrays and spectral processing (e.g., YAMNet), present reduced costs but face limitations in range and sensitivity to background noise and the aircraft's own motors [11, 12].

1.2.6 Sensor Comparison Table

Table 1 consolidates the technical parameters that underpinned the selection of the hybrid architecture for Point D.

Table 1: Technical Comparison Matrix of Sensors for ABDAA

Criterion	**LiDAR** *(High-End)*	**mmWave Radar** *(TI IWR6x)*	**Comp. Vision** *(Edge AI)*	**Acoustic** *(Passive)*
Consumption (W)	14–20	1.5–2.5	5–15	2–3
Weight (g)	430–500	<10	20–100	50–150
Computational Cost	Medium	Low	Very High	Low
Latency (ms)	<10	25–50	12–40	100–500
Weather Resilience	Low	Very High	Medium	Medium
Typical Range (m)	35–100	150–300	500–1200	100–300

This matrix justifies the path taken: the SIMUA project architecture prioritizes the fusion between mmWave Radar (for robustness and proximity) and Computer Vision (for long-range detection), discarding LiDAR due to its severe impact on the operational envelope of small drones.

1.2.7 International References and Empirical Parameters

The definition of input parameters for Point D simulations requires a solid foundation in empirical and normative data. Detection envelopes establish the minimum safety distances and warning times necessary for the DAA system to execute effective evasive maneuvers [13].

Technical literature, including studies from NASA, Amazon Prime Air, and LFV, defines realistic detection distances that vary according to the technology and the radar cross section (Radar Cross Section (RCS)) or visual signature of the intruder:

- **Computer Vision**: Studies conducted by NASA in AAM test environments indicate 96.1% to 99.9% coverage for octocopter-type aircraft (e.g., Alta 8) at distances greater than 1 km [14]. Angular precision is superior to distance precision (*range*), introducing uncertainties handled by the offset algorithm.

- **mmWave Radar**: Compact embedded systems operating at 60 GHz (such as the IWR6843) have optimized ranges of 150 to 300 m, potentially reaching 1.2 km for micro-UAVs in high-sensitivity configurations, although they face challenges with ground *clutter* at low altitudes [15].

- **Time Thresholds**: For compatibility with air traffic control (Air Traffic Control (ATC)), the system must initiate separation maintenance maneuvers (*Well Clear*)

between 60 and 90 seconds before the closest point of approach (Closest Point of Approach (CPA)) [16].

1.2.8 Probability of Detection Modeling

The probability of detection (P_d) is a dynamic factor that introduces realism to the simulation. It is not only a function of distance but also of stochastic environmental variables:

- **Meteorological Impact**: In mmWave radars, attenuation by heavy rain can reduce the effective range by 20–30%, while computer vision can reach $P_d \to 0$ in total fog or absolute darkness [17].
- **Background Acoustic Noise**: P_d drops exponentially with distance. Microphone arrays with advanced processing can extend detection from 200 m to up to 500 m, but with a high false positive rate in urban areas [18].
- ***Clutter* and Geometry**: The probability of maintaining a *firm track* via radar drops to levels between 36% and 56% in critical low-altitude scenarios with signal multipath [14].

1.2.9 Detection Envelopes Table

Table 2 consolidates the detection envelopes adopted for SIMUA modeling.

Table 2: Detection Envelopes and Operational Thresholds

Sensor	R_{max} **(m)**	T_{warn} **(s)**	**Uncertainty (m)**	**Critical Factor**
ADS-B (Cooperative)	>9,000–18,500	90–120	<5	MSO channel saturation
Computer Vision	800–1,200	45–60	10–20	Fog / Lighting
mmWave Radar	150–300	15–30	2–5	*Ground clutter* / RCS
Acoustic (Passive)	100–300	10–20	>20	Wind / Own motors

2 Experimental Campaign and Performance Results

2.1 Macro-Scale Strategic Layering Analysis (MSLA)

(Formerly "testeprimordial" series)

This section documents a series of large-scale simulations conducted in Python to establish the conceptual layered model of conflict management. The MSLA emphasizes the strategic layering of UTM (UAS Traffic Management Extension (xTM)) pre-departure deconfliction and tactical DAA (RWC) as complementary safety barriers. The experiments simulate 8 hours of drone operations within a geofenced area centered in São José dos Campos (SJC), using geodesic lat/lon coordinates and *spatial hashing* for high-density performance.

2.1.1 Scenario 1 — Baseline: Blind Flight

The purpose of this scenario is to establish a safety baseline by measuring near-collisions when drones operate without awareness of each other — without DAA and without centralized UTM (xTM) management.

- **Operational Model**: Drones follow random origin/destination paths within the SJC geofence, cruising at altitudes of 30–50 m AGL.
- **Safety Metric**: Measured via *MACproxy* (unique pairs of drones violating the separation of 20 m horizontal and 10 m vertical).
- **Key Results**: The experiment generates a "chaos curve" illustrating the exponential increase in near-collisions as drone density scales from 5 to 150 aircraft.

2.1.2 Scenario 2 — DAA Only (Reactive)

This scenario evaluates how an isolated reactive DAA layer affects safety and route efficiency without strategic management.

- **Technique**: Drones scan for threats within a 150 m radius and execute a tangential evasion for 8 seconds.
- **Observations**: Although DAA significantly reduces collisions, it introduces route inefficiency. The system "collapses" (inefficiency >10%) at specific density thresholds.

2.1.3 Scenario 3 — xTM Only (Strategic)

Evaluates capacity when using centralized 4D trajectory reservations as the sole separation mechanism.

- **Technique**: The `xTMCentral` class performs prior deconfliction of 4D tubes (30 m H / 15 m V). Rejected drones wait in 15-second retry slots.
- **Observations**: This model eliminates collision risk by design but increases ground delay as the airspace reaches its capacity.

2.1.4 Scenario 4A — Optimized xTM + DAA

Demonstrates that a tactical DAA layer allows for narrower xTM corridors, increasing total throughput.

- **Strategy**: xTM uses reduced separation tubes (22 m H / 12 m V), relying on DAA to handle minor tactical deviations.
- **Impact**: Significant increase in sustainable density before reaching the 5-minute average takeoff delay threshold.

2.1.5 Scenario 4B — xTM + Intelligent DAA + Stochastic Wind (Encounter Model Validation)

This scenario serves as a validation framework using *Encounter Models*, the standard method for assessing DAA safety levels [19]. The stochastic wind perturbation generates both *correlated* encounters (aircraft on similar routes disturbed into conflicting paths) and *uncorrelated* encounters (random crossings), enabling a statistically representative evaluation of the RWC function.

- **Wind Model**: Stochastic perturbation of ± 1.5 m/s in cruise speed, inducing trajectory deviations that produce encounter geometries spanning head-on, crossing, and overtaking configurations.
- **Encounter Types**: Correlated encounters arise from fleet operations on adjacent corridors perturbed by wind; uncorrelated encounters arise from independent random origin-destination pairs.
- **DAA Update**: Intelligent avoidance logic (RWC layer) that determines the ideal side of evasion (left/right) based on relative bearing.
- **Results**: Proves that elastic reservations with time buffer, combined with tactical DAA, maintain safety even under unpredictable flight conditions. The encounter-based framework enables direct comparison with the safety levels prescribed by ASTM F3442 [2].

2.1.6 Summary of Python Experiments

Table 3 summarizes the features and primary metrics of the *testeprimordial* series.

Table 3: Comparison of Python Simulation Scenarios

Aspect	Scenario 1	Scenario 2	Scenario 3	Scenario 4A	Scenario 4B
xTM	No	No	Yes (30/15 m)	Yes (22/12 m)	Yes (Elastic)
DAA	No	Reactive	No	Tactical	Intelligent
Wind	No	No	No	No	Yes ($\pm$ 1.5 m/s)
Restricted Area	No	No	Yes	Yes	Yes
Primary Metric	MACproxy	Inefficiency	Avg. Delay	Avg. Delay	Avg. Delay
Auto-stop	None	Ineff $\geq$ 10%	Delay $\geq$ 5 min	Delay $\geq$ 5 min	Delay $\geq$ 5 min

2.2 High-Fidelity Tactical Logic Validation (HTLV)

(Formerly Rust/DAIDALUS Integration)

Following the MSLA conceptual experiments, this section presents the HTLV: a high-fidelity simulation developed in Rust, which integrated NASA's DAIDALUS as the standards-compliant RWC engine. The HTLV validates compliance with RTCA DO-365B [1] and ASTM F3442 [2] performance requirements through precision kinematic updates at 50 Hz.

2.2.1 Comparison: Python vs. Rust/DAIDALUS

The comparison in Table 4 highlights the complementary strengths of the two modeling approaches.

Table 4: Python Experiments vs. Rust/DAIDALUS Simulation

Dimension	**Python (testeprimordial)**	**Rust / DAIDALUS Simulation**
Coordinate System	Geodetic Lat/Lon (SJC)	Cartesian XYZ (5 × 5 km box)
Physics Engine	Simple kinematics (dt = 1 s)	Bevy Entity Component System (ECS) + Rapier3D (50 Hz)
DAA Engine	Custom heuristic (proximity-based)	NASA DAIDALUS (DO-365B)
Alert Information	Binary (Threat / No Threat)	Multi-level (1-3), TTV, Heading Bands
Avoidance Maneuver	Fixed angular (60–90°)	DAIDALUS Bands + Horiz./Vert. offsets
Scale	High (150+ drones, 8h runs)	Medium (200 drones, controlled tests with 2 drones)
Visualization	Plots + Matplotlib	WebGL Interactive 3D Viewer

2.2.2 Complementary Strengths and Integration Path

The **Python experiments** effectively validate the three-layer conflict management model defined in the CONOPS — Strategic (xTM), Tactical (DAA), and Collision Avoidance. They utilize real SJC geography and metrics directly applicable to DECEA assessment.

The **HTLV (Rust/DAIDALUS)** provides the fidelity necessary to demonstrate compliance with international standards such as RTCA DO-365B [1] and ASTM F3442 [2]. Its high-frequency physics engine (ECS at 50 Hz) ensures precise kinematic updates during RWC avoidance maneuvers, and the alerting logic incorporates HITL latency margins (5–15 s) for RPIC decision authority.

2.2.3 Path Ahead: Integration Proposal

One of the proposed paths for a unified demonstration to DECEA is that the Rust/-DAIDALUS simulation telemetry be fed into the Python *MACproxy* analysis pipeline. This will allow the calculation of the *MACproxy* "chaos curve" using a standard-compliant DAA engine, thus demonstrating the maximum capacity of the SJC urban corridor.

2.2.4 Summary for DECEA Evaluation

The combined value chain — from 4D strategic reservations in Python to standard-compliant tactical deviation in Rust — demonstrates a robust approach for managing UAS traffic in non-segregated Brazilian airspace. This integrated model provides a complete narrative, from identifying the "Blind Flight" baseline problem to validating a high-fidelity interoperable solution.

2.3 DAIDALUS Core Logic and Multi-Sensor Fusion

2.3.1 DAIDALUS as the RWC Implementation

The DAIDALUS (*Detect and Avoid Alerting Logic for Unmanned Systems*) algorithm, developed by NASA, constitutes the logical backbone of Point D (*Architecture Demonstration*) in the SIMUA project [20]. Within the DAA layered architecture, DAIDALUS implements the RWC function as defined in RTCA DO-365B [1]: it transforms surveillance data into maneuver guidance (*Bands*) for the RPIC or the autonomous control system, with the objective of maintaining the Well Clear separation volume. The CA layer (emergency collision avoidance) acts as a last-resort barrier only when the RWC function has failed to maintain separation.

2.3.2 Base Scenario: Cooperative Encounters via ADS-B/Telemetry

In the cooperative scenario, DAIDALUS operates with high-fidelity telemetry data, with aircraft communicating via ADS-B transponders or UTM data links [21]. The algorithm utilizes the *Well-Clear* metric, defined by a cylindrical or ellipsoidal protection volume around the aircraft [22].

The mathematical logic for the Well Clear Violation (WCV) alert is based on time and distance inequalities:

$$\tau_{mod}(s, v) < \text{TTHR} \quad \wedge \quad s \cdot v < 0 \quad \wedge \quad \|s_H\| < \text{DMOD} \tag{1}$$

where τ_{mod} is the modified time to convergence, s is the relative position vector, v is the relative velocity vector, DMOD is the minimum vertical separation distance, and TTHR is the warning time threshold [22].

In cooperative scenarios, the resolution bands suggested by DAIDALUS are precise, allowing aircraft to maintain separation with minimal course deviations, maximizing traffic flow efficiency.

2.3.3 Advanced Scenario: Non-Cooperative Aircraft with Onboard Sensors

The greatest technical challenge, and the strategic differentiator of this report, is the behavior of the DAA logic when fed by onboard sensors (Radar and Vision) that detect non-cooperative aircraft. Unlike ADS-B, these sensors provide "low-fidelity" data, characterized by noise and systematic measurement errors [23].

2.3.4 Sensor Uncertainty Mitigation (SUM)

To handle sensor imprecision, version 2 of DAIDALUS incorporates SUM (*Sensor Uncertainty Mitigation*) logic [24]. The SUM framework is critical for proving that the RWC function maintains a safe Well Clear volume even when fed by “low-fidelity” data from

computer vision and mmWave radar. Instead of considering the intruder's position as a single point, SUM uses error distributions (*z-scores*) to create dynamic safety *buffers*:

- **Buffer Adjustment**: If a visual detection via YOLOv10 reports a drone at 500 m with an estimated error of 5 m, DAIDALUS SUM expands the conflict volume to encompass this uncertainty, ensuring that the evasive maneuver is calculated based on the intruder's "worst probable position" [23]. This accounts for sensor noise, measurement latency, and systematic biases inherent in onboard perception.
- **Low Fidelity *vs.* Telemetry**: While precise telemetry data allow for smooth and late maneuvers, noisy onboard sensor data require earlier alerts and more conservative maneuvers. The system compensates with increased horizontal and vertical uncertainty parameters (s_ϵ, v_ϵ, sz_ϵ, vz_ϵ), enhancing safety at the cost of a moderate increase in pilot workload [23].
- **Latency Compensation**: Sensor processing delays (12–50 ms for vision/radar) are incorporated into the SUM parameters, ensuring that the effective Well Clear volume accounts for the intruder's predicted motion during the processing pipeline.

2.3.5 Virtual Intruder State Aggregation (VISA)

Complementary to SUM, the VISA method aggregates multiple subsequent intruder state observations over a sliding time window to produce a smoothed trajectory estimate. This reduces the effect of measurement *jitter* on the maneuver bands and prevents erratic flight commands that would otherwise result from frame-to-frame noise in computer vision detections or radar returns [23]. By averaging position and velocity estimates across consecutive sensor updates, VISA provides DAIDALUS with a stable intruder state vector, ensuring that the RWC guidance bands remain consistent and actionable for the RPIC.

2.3.6 Human-in-the-Loop (HITL) Alerting Considerations

The DAIDALUS alerting thresholds are configured to account for RPIC decision authority and response latency. In remote piloting scenarios, the communication and cognitive processing chain introduces a latency of 5–15 seconds between alert presentation and maneuver initiation [16]. The three-level alerting structure is designed to accommodate this HITL factor:

- **Preventive Alert** (60 s to CPA): Provides early awareness, allowing the RPIC to assess the situation and plan a response within the available decision window.
- **Corrective Alert** (30 s to CPA): Requires immediate action; the 5–15 s pilot latency is absorbed within the remaining time margin.

- **Warning Alert** (15 s to CPA): Triggers autonomous or semi-autonomous evasion if the RPIC has not responded, transitioning toward the CA layer.

These parameters ensure that the alerting timeline provides sufficient margin for human decision-making while maintaining the integrity of the RWC function.

2.3.7 Simulation Environment

The simulations were conducted in the following environments:

- **Colosseum (AirSim)**: High-fidelity simulator for flight tests in urban and rural scenarios, with support for synthetic sensor data generation [25].
- **DJI Simulator**: Used for validating flight parameters on small commercial platforms.
- **DAIDALUS v2 (NASA)**: Run as an integrated software module in the decision *pipeline*, receiving aircraft states in a standardized format [26].

2.4 Integrated Detection and Resolution Performance

2.4.1 Detection Performance

- **Accuracy Metrics**: Trained models (based on Det-Fly and AOT for computer vision) achieved mAP (*mean Average Precision*) exceeding 85% under standard lighting conditions and low traffic density.
- **Mean Detection Distance**:
  - mmWave Radar: 210 m (median);
  - Computer Vision: 650 m (normal lighting).
- **Impact of Variable Conditions**: Computer vision shows mAP degradation of approximately 30–40% under fog (visibility <500 m) and 60–70% in dark conditions without infrared.

2.4.2 Conflict Resolution Effectiveness

- **DAIDALUS RWC Encounters**: In simulations with the DAIDALUS algorithm active, 100% of conflict scenarios detected >15 s from the CPA were resolved without violating the Well Clear volume.
- **Without DAIDALUS (Blind Flight Baseline)**: In control scenarios (without algorithmic intervention), the minimum separation violation rate was $X\%$.

- **Miss Distance**: The minimum separation distance maintained in simulations supervised by DAIDALUS was consistently greater than $D_{WC} = 1{,}500\,\text{ft}$ horizontal / 450 ft vertical (standard DAIDALUS parameters for UAS).

2.4.3 Risk Ratio Analysis

The *Risk Ratio* quantifies the safety improvement provided by the DAA system by comparing the probability of a NMAC with the system active versus the "Blind Flight" baseline (Scenario 1 of the MSLA):

$$R_{\text{ratio}} = \frac{P(\text{NMAC} \mid \text{DAA active})}{P(\text{NMAC} \mid \text{Blind Flight})} \tag{2}$$

A Risk Ratio $\ll 1$ demonstrates that the ABDAA architecture provides a quantifiable reduction in collision risk. In the MSLA simulations, the combined xTM + intelligent DAA configuration (Scenario 4B) reduced the MACproxy count to near-zero at densities where the Blind Flight scenario exhibited exponential conflict growth, yielding a Risk Ratio on the order of 10^{-2} at moderate densities (5–7 UAS/km^2).

2.4.4 Nuisance Alert Analysis

Nuisance alerts — alerts triggered when no actual threat exists or when the conflict resolves naturally without intervention — are a critical metric for operational efficiency and RPIC workload [16]. A high nuisance alert rate degrades pilot trust in the system and increases cognitive load, potentially leading to alert fatigue.

- **Cooperative Scenario**: With high-fidelity ADS-B data, nuisance alert rates were minimal, as the precise state information allows DAIDALUS to accurately predict conflict geometry and suppress alerts for non-threatening encounters.
- **Non-Cooperative Scenario (SUM enabled)**: The expanded uncertainty volumes from SUM logic increase the preventive alert rate. However, the VISA trajectory smoothing reduces false transitions between alert levels, keeping the corrective and warning alert nuisance rate within acceptable bounds.
- **Operational Impact**: Nuisance alert rates are monitored as a function of sensor fidelity and encounter geometry to ensure that the system remains operationally viable without inducing alert fatigue in the RPIC.

2.4.5 Datasets and AI Models Used

- **Det-Fly Dataset**: Used for training the in-flight aircraft visual detector, contains images of drones in various background and lighting scenarios.

- **AOT (*Associating Objects with Transformers*)**: Employed for multi-object tracking, maintaining the identity of detected targets between consecutive frames.
- **YOLOv8 / YOLOv10**: Real-time detection architectures, run on NVIDIA Jetson Orin Nano or Google Coral Edge TPU hardware, depending on the platform configuration.
- **DAIDALUS Integration**: Visual detections are converted into state vectors (estimated position, relative velocity, uncertainty) and injected into DAIDALUS via a standardized message interface, with SUM logic enabled to handle distance uncertainty.

3 Operational Impact and Airspace Capacity

3.1 Airspace Saturation Analysis

3.1.1 Stress Test Methodology in Urban Corridors

Area saturation analysis transforms technology into operational policy and validates the *Scalability* requirement of the SIMUA CONOPS. Through *Stress Tests*, an urban delivery corridor is simulated, and drones are progressively added until the system reaches its capacity limit [27]. The objective is to demonstrate that the ABDAA architecture enables UAS densities exceeding 7 UAS/km^2 in urban corridors — the threshold identified in the CONOPS as the minimum capacity required for commercial viability of urban drone delivery.

The simulation uses a multi-commodity network flow (Multi-Commodity Network Flow (MCNF)) model to manage routes and departure times, ensuring that temporal separation is maintained at all nodes and edges of the airspace graph [28]. Capacity is measured in terms of maximum flow rate (*Throughput*), the probability of unresolved conflicts, and the Risk Ratio (Eq. 2) at each density level.

3.1.2 Capacity Comparison: The Impact of DAA Architecture

Saturation simulation results demonstrate the differential value of each architecture:

1. **Without DAA (Blind Flight)**: The area saturates at extremely low density. Safety depends on massive geographic *buffers* and rigid temporal separation, which, as established in the SIMUA CONOPS, is inherently unscallable for the projected volume of UAS and eVTOL operations in urban centers.

2. **With DAIDALUS (Cooperative)**: Capacity increases significantly. The use of ADS-B or remote identification (Remote Identification (RID)) allows drones to "see each other" for kilometers, enabling optimized crossings and alignment with the Federated Services Network architecture. However, the system encounters physical limits in the communication channel capacity (e.g., MSO slot collisions) as density increases.

3. **With Hybrid ABDAA (Sensors + DAIDALUS)**: This architecture enables the highest sustainable density (> 7 UAS/km^2), directly validating the Scalability requirement of the CONOPS. By combining global management (UTM) with autonomous local perception (Radar/Vision), the system overcomes digital “blind zones” and facilitates the transition from isolated corridors to a high-capacity, scalable integrated framework. The RWC function ensures tactical safety at densities where strategic deconfliction alone is insufficient.

3.1.3 Quantitative Saturation Results

Table 5: Airspace Capacity Analysis and Saturation in Urban Environments

Architecture	**Density (UAS/km²)**	P_{decode} **ADS-B**	**Conflict Risk**	**Throughp (Relativ**
Standard VFR (Without DAA)	1.0 (Low)	N/A	High	100% (Bas
Cooperative (ADS-B 10mW)	3.0 (Medium)	0.84–0.95	Low	250%
Cooperative (High Density)	5.0 (High)	0.28–0.68*	Critical	150%†
SIMUA Hybrid (ABDAA)	>7.0 (Very High)	N/A (Local Sensor)	Minimum	>400%

*Probability degraded by MSO slot collisions of the ADS-B protocol.

†Throughput degraded relative to the medium density scenario.

3.2 Alignment with CONOPS and SISCEAB

3.2.1 Validation of Operational Premises

The results of Point D confirm the strategic framework established in the SIMUA project CONOPS (OCD v0). The hybrid ABDAA architecture addresses the limitations identified in the current air traffic management system, which depends on large buffer zones and manual coordination. By implementing autonomous local perception with a standards-compliant RWC function (DAIDALUS/DO-365B), the architecture supports the CONOPS vision of transitioning from static airspace blocks to a flexible, high-capacity environment.

3.2.2 Link to SISCEAB Strategic Objectives

The technical results directly support the strategic mandates of DECEA and Agência Nacional de Aviação Civil (ANAC) for airspace modernization:

- **Capacity Expansion**: The saturation analysis (Section 3.1) demonstrates that the ABDAA architecture enables densities > 7 UAS/km^2, directly supporting DECEA's objective of transitioning from static airspace blocks to flexible, high-capacity environments that accommodate both manned and unmanned operations.

- **Operational Flexibility**: By combining strategic xTM deconfliction with tactical RWC self-separation, the architecture allows DECEA to incrementally open airspace classes (G $\rightarrow$ E $\rightarrow$ D $\rightarrow$ C) to UAS operations without requiring permanent segregated corridors.

- **Sovereign Integration**: The simulation framework, validated with SJC geography and SISCEAB airspace class definitions (per the OSED in Section 1.1), provides DECEA with a nationally developed capability for evaluating DAA performance under Brazilian operational conditions.

3.2.3 Compliance with SISCEAB and Federated Management

As defined in the project scope, the demonstrated architecture integrates with the Brazilian Airspace Control System (SISCEAB) while fulfilling the requirements for a decentralized and federated UTM model. It meets the following national and international milestones:

- **Delegated Responsibility**: The CONOPS requires that tactical deconfliction be strategically delegated from ATC to UAS operators or the RPIC. The use of DAIDALUS ensures the technical feasibility of this transfer of responsibility.
- **Support for DECEA**: *Well Clear* parameters are synchronized with Brazilian guidelines for BVLOS in Classes G and E, supporting the goal of sovereign airspace integration.
- **ANAC Principle of Equivalence**: The hybrid approach satisfies safety equivalence for operations in populated areas (RBAC-E 91) without the need for permanent segregated areas.

3.2.4 Mitigation of Identified Risks

Table 6 maps the collision risks identified in the CONOPS safety analysis — including *Geographical and Jurisdictional* risks — and the mitigations provided by the ABDAA architecture. Each mitigation is directly linked to a specific ABDAA feature, demonstrating traceability from risk to technical solution.

Table 6: Risk Mapping and Mitigation by the ABDAA Architecture

Identified Risk	**Scenario**	**Mitigation (ABDAA Feature)**	**CONOPS Category**	**Res.**
Collision with non-cooperative UAS	Urban BVLOS	mmWave Radar Fusion + SUM/VISA (RWC)	Operational	Low
ADS-B channel failure	High density	Autonomous local perception (*backup* sensor)	Technical	Low
Low-light / fog detection	Night / Adverse Weather	mmWave Radar (weather-immune) + IR Fusion	Geographical	Medium
Low-altitude *ground clutter*	Near urban structures	Adaptive FMCW filtering + CFAR	Geographical	Medium
Edge processor saturation	High traffic density	Distance prioritization (*Near-Field gate*)	Technical	Low
RPIC response latency	All BVLOS	HITL-aware alerting thresholds (5–15 s margin)	Human Factors	Low
Jurisdictional airspace transition	Cross-class boundary	OSED-defined parameter switching (G/E/D/C)	Jurisdictional	Low

4 Conclusions and Strategic Vision

4.1 Conclusions and Strategic Vision

4.1.1 Technical Feasibility of the Proposed Architecture

The methodology presented in this report consolidates the technical transition necessary for the delivery of Point D of the ITA/ICEA project. The dual-scale experimental campaign — MSLA for strategic layering validation and HTLV for standards-compliant tactical logic — demonstrates that the choice of components and algorithmic logic are based on a deep understanding of the physical and operational constraints of Brazilian airspace:

- **Investment Justification**: The exhaustive proof that LiDAR is unsuitable for sUAS, due to SWaP constraints, validates the strategic need to invest in solid-state sensors (mmWave) and high-efficiency Computer Vision (*Edge AI*).
- **RWC Compliance**: The HTLV campaign demonstrates that DAIDALUS, configured as a RWC implementation per DO-365B [1], can be effectively paired with SUM and VISA logic to handle low-fidelity data from noisy onboard sensors. This removes one of the greatest obstacles to the integration of UAS into non-segregated airspaces: the uncertainty of onboard surveillance data.

- **Safety and Capacity**: The MSLA saturation study proves that the Hybrid ABDAA architecture enables densities > 7 UAS/km^2, validated through encounter models with correlated and uncorrelated geometries [19]. The Risk Ratio analysis confirms orders-of-magnitude safety improvement over the Blind Flight baseline.
- **HITL Viability**: The alerting timeline, designed with 5–15 s RPIC latency margins, ensures that the system remains operationally viable under human-in-the-loop decision authority, with nuisance alert rates controlled by VISA trajectory smoothing.

4.1.2 Identified Limitations

1. **Extreme Lighting Conditions**: Computer vision presents severe limitations in nocturnal scenarios without infrared, requiring sensory redundancy (Radar + IR).
2. **Ground Clutter**: mmWave Radar is affected by multipath near urban structures at low altitudes, requiring adaptive filtering algorithms.
3. **Acoustic Latency**: The passive acoustic sensor, with latency of 100–500 ms, is unsuitable as a primary DAA sensor for rapid approach scenarios.
4. **ADS-B Range in High Density**: MSO protocol saturation limits the reliability of ADS-B in densities exceeding 5 UAS/km², making onboard sensors indispensable.

4.1.3 Contribution to the SIMUA Project and Future Vision

The results of Point D directly contribute to the SIMUA project's international mission by providing:

- A replicable technical base for AAM integration scenarios in Brazilian cities (São Paulo, Campinas, São José dos Campos) and Swedish cities (Stockholm, Gothenburg), with an OSED mapped to SISCEAB airspace classes (G, E, D, C).
- A dual-scale simulation framework — MSLA (Python/strategic) and HTLV (Rust/-tactical) — extendable to commercial delivery, surveillance, and emergency response, with standards compliance verified against RTCA DO-365B and ASTM F3442 [1, 2].
- Foundations for the development of national standards for certifying ABDAA systems, with quantitative safety metrics (Risk Ratio, nuisance alert rate, throughput) aligned with ASTM F3442, EUROCAE ED-269, and RTCA SC-228 benchmarks.

The supporting media attached to this report — recordings of saturation simulations and DAIDALUS behavior logs under stress — serve as empirical proof of the modeled algorithms' effectiveness and underpin the certification requirements and RWC/CA strategies that will enable the harmonious and safe coexistence between manned and unmanned aircraft in national airspace.

5 UTM Simulation Scenarios

To evaluate the operational capacity and safety limitations within dense urban environments, a set of computational simulation scenarios was designed for the Unmanned Aircraft System Traffic Management (UTM) and Advanced Air Mobility (AAM) airspace in São José dos Campos. The modeling relies on parameters derived from international standards and the current Concept of Operations.

5.1 Basic Simulation Parameters

The simulations consider a nominal cruise speed of 15.3 m/s for VTOL-type delivery drones, operating at an altitude range of 30.0 to 50.0 meters. Vertical maneuvers for takeoff and landing are performed at 3.0 m/s. The maximum mission distance parameter is set to 12.0 km to respect usual Small UAS (sUAS) battery restrictions.

For safety evaluation, the *Mid-Air Collision Proxy* (MACproxy) is defined as a critical separation failure, utilizing a cylindrical volume of 20.0 m in radius (horizontal) and 10.0 m in height (vertical). These dimensions are in accordance with the ASTM F3442-25 guidelines.

The architecture was evaluated across five distinct operational scenarios, incrementally adding layers of Strategic (xTM) and Tactical (DAA) management.

5.2 Scenario 1: UTM without xTM and without DAA

This baseline scenario simulates unmanaged free-flight traffic where no preliminary 4D strategic allocation is performed prior to takeoff, and vehicles lack autonomous tactical avoidance (DAA) capabilities.

Justification: It serves as a control scenario to demonstrate the severe inefficiency and high collision rates of unstructured traffic in dense urban areas, confirming the necessity of a structured UTM approach and emphasizing the baseline MACproxy intrusion rates.

5.3 Scenario 2: UTM with DAA only

In this setup, strategic management (xTM) is absent, meaning flights take off without guaranteed safe paths. However, the drones are equipped with an autonomous Detect and Avoid (DAA) system operating with a simulated sensor range of 25.0 to 150.0 m horizontally and 12.0 to 30.0 m vertically.

Justification: This scenario tests a purely decentralized tactical deconfliction architecture. It evaluates whether local embedded sensors alone are sufficient to maintain safe separation under dense traffic conditions without the macro-structural organization of pre-flight flow management.

5.4 Scenario 3: UTM with xTM only

This scenario relies purely on a centralized, strategic management approach. Every flight must secure an approved 4D tube (with 22.0 to 50.0 m horizontal and 12.0 to 20.0 m vertical safety buffers) prior to departure. The drones do not possess the ability to perform tactical maneuvers during flight. Operational limits dictate a maximum ground delay of 5.0 minutes and a logistic saturation cap where route inefficiency cannot exceed 10.0%.

Justification: It assesses the effectiveness of purely strategic flow management and Demand-Capacity Balancing (DCB). By measuring the rate of denied slots and logistic saturation, this test quantifies the limitations of an architecture that lacks the flexibility of tactical en-route adjustments.

5.5 Scenario 4: UTM with xTM and DAA

This setup implements the complete hybrid safety architecture proposed in the project. The xTM layer strategically allocates 4D tubes to resolve macro-conflicts before departure, while the DAA system remains active during the flight to tactically resolve residual or unforeseen conflicts.

Justification: This is the state-of-the-art approach representing the target UTM environment. It combines centralized planning for efficiency with decentralized tactical execution to guarantee absolute flight safety. The scenario evaluates the combined synergy and overall throughput optimization.

5.6 Scenario 4b: UTM with xTM, DAA, and Wind Variations

Building upon the robust hybrid architecture of Scenario 4, this variant introduces severe kinematic uncertainty. It applies stochastic wind gusts that alter the nominal speed of the drones by $\pm$ 1.5 to 2.0 m/s.

Justification: The purpose of this stress-test is to evaluate the resilience of the 4D tube model and the calculated takeoff times (CTOT) against realistic urban wind shear and weather phenomena. It determines if the combined strategic and tactical layers can maintain safety margins without exceeding the 10.0% logistic inefficiency threshold when kinematic deviations frequently trigger the DAA layer.

.1 Simulation Logs

Raw simulation data files (CSV/JSON) are available in the project repository:

- `sim_cooperative_daidalus.csv`: Logs for the cooperative scenario (ADS-B + DAIDALUS).
- `sim_noncooperative_sum.csv`: Logs for the non-cooperative scenario (SUM/VISA enabled).
- `saturation_stress_test.json`: Saturation *stress test* data.

.2 Simulation Video Gallery and Captures

- **Video 1**: Visual detection via YOLOv10 + activation of DAIDALUS bands (`demo_visual_daa.mp4`)
- **Video 2**: Non-cooperative drone tracking via mmWave Radar (`demo_mmwave_track.mp4`)
- **Video 3**: Saturation scenario – urban corridor with >7 UAS/km^2 (`saturation_corridor.mp4`)

.3 DAIDALUS v2 Configuration Parameters

The parameters below represent the configuration used in the Point D simulations. They can be loaded directly by DAIDALUS from a `.conf` file.

```
# ==============================================================
# DAIDALUS v2 - SIMUA Point D Configuration (UAS-to-UAS)
# Instituto Tecnológico de Aeronáutica / ICEA
# ==============================================================

# --- Well Clear Volume (UAS-to-UAS) ---
DTHR  = 1500 ft       # Horizontal Distance Threshold
TTHR  = 35 s          # Time Threshold (modified tau)
ZTHR  = 450 ft        # Vertical Separation Threshold
TCOA  = 0 s           # Time to Co-Altitude

# --- Sensor Uncertainty Mitigation (SUM) ---
s_EPS  = 0.5          # horizontal position uncertainty (NMI)
v_EPS  = 0.5          # horizontal velocity uncertainty (kt)
sz_EPS = 150 ft       # vertical position uncertainty
vz_EPS = 50 fpm       # vertical rate uncertainty
```

```
# --- Band Resolution ---
turn_rate = 1.5 deg/s # maximum turn rate for resolution
vs_step   = 100 fpm   # vertical speed band step
gs_step   = 15 kt     # horizontal speed band step

# --- Alerting Thresholds ---
alert_1_time   = 60 s   # Preventive alert
alert_2_time   = 30 s   # Corrective alert
alert_3_time   = 15 s   # Warning alert
```

»»»> f6a52f605c711f640dc69baf6e905eb6e59a5860

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List of Acronyms and Abbreviations

AAM Advanced Air Mobility.

ABDAA Airborne Detect and Avoid.

ADS-B Automatic Dependent Surveillance-Broadcast.

ANAC Agência Nacional de Aviação Civil.

ATC Air Traffic Control.

BVLOS Beyond Visual Line of Sight.

CA Collision Avoidance.

CONOPS Concept of Operations.

CPA Closest Point of Approach.

DAA Detect and Avoid.

DECEA Departamento de Controle do Espaço Aéreo.

ECS Entity Component System.

FMCW Frequency Modulated Continuous Wave.

HITL Human-in-the-Loop.

HTLV High-Fidelity Tactical Logic Validation.

ICEA Instituto de Controle do Espaço Aéreo.

ITA Instituto Tecnológico de Aeronáutica.

LFV Luftfartsverket (Swedish Civil Aviation Authority).

MCNF Multi-Commodity Network Flow.

MSLA Macro-Scale Strategic Layering Analysis.

NMAC Near Mid-Air Collision.

OSED Operational Services and Environment Definition.

RCS Radar Cross Section.

RID Remote Identification.

RPIC Remote Pilot in Command.

RWC Remain Well Clear.

SISCEAB Brazilian Airspace Control System.

sUAS Small Unmanned Aircraft System.

SUM Sensor Uncertainty Mitigation.

SWaP Size, Weight and Power.

UAS Unmanned Aircraft System.

UTM Unmanned Traffic Management.

VISA Virtual Intruder State Aggregation.

WCV Well Clear Violation.

xTM UAS Traffic Management Extension.